DEGREE CERTIFICATE DETAIL EXTRACTION

(B. Tech. Project-2)

A REPORT

Submitted by

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for the partial fulfillment of the requirements for Semester – VII of

BACHELOR OF TECHNOLOGY

(INFORMATION TECHNOLOGY)

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November 2026

Candidate Disclosure on the Use of AI Tools

In the process of writing this project report, developing software modules, and conducting model experiments, the following artificial intelligence (AI) tools and technologies were utilized responsibly and ethically:

- **Antigravity / Gemini (3.8 Flash) & Claude (Sonnet 4.6):** Used for code refactoring, model training pipeline debugging, optimizing GPU tensor operations, automating repetitive boilerplate logic for bounding box calculations, and assisting in structured technical documentation and formatting.
- **Claude (Sonnet 5):** Used during the initial exploratory research phase for architectural ideation, comparative trade-off analysis between OCR-dependent (e.g., LayoutLM) and OCR-free (e.g., Donut) document understanding architectures, dataset augmentation strategies, and evaluation methodology design.

All final system designs, mathematical formulations, experimental verifications, dataset curations, code implementations, model evaluations, and written interpretations in this report represent the authentic intellectual work, implementation, and academic findings of the candidate.

Candidate's Declaration

I declare that the dissertation (for B. Tech in Information Technology) titled "Degree Certificate Detail Extraction" is my own work being conducted under the guidance and supervision of Dr. Harshadkumar B. Prajapati, Professor, Department of Information Technology, Dharmsinh Desai University.

I further declare that to the best of my knowledge, this dissertation does not contain any part of work which has been submitted for the award of any degree either in this University or in any other University without proper citation.

Signature: _____

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CERTIFICATE

This is to certify that this Report of B. Tech. Project-2 titled “Degree Certificate Detail Extraction” submitted for partial fulfillment of B. Tech Semester-VII is a record of the genuine project work carried out by:

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Acknowledgment

First and foremost, I express my profound gratitude and heartfelt thanks to my project guide, Dr. Harshadkumar B. Prajapati, Professor, Department of Information Technology, Dharmsinh Desai University, for his invaluable guidance, continuous encouragement, constructive criticism, and technical insights throughout the course of this project. His inspiring mentorship has played a pivotal role in the successful formulation and execution of this research.

I extend my sincere thanks to Prof. Dr. V. K. Dabhi, Head of the Department of Information Technology, for providing the necessary computing resources, departmental infrastructure, and academic environment conducive to advanced research in deep learning and computer vision.

I also take this opportunity to thank all the faculty members and technical staff of the Department of Information Technology, Faculty of Technology, Dharmsinh Desai University, who directly or indirectly aided and encouraged me during this endeavor.

Lastly, I express my deepest gratitude to my parents and family for their unwavering moral support, patience, and sacrifices throughout my academic journey, and to my peers for their valuable discussions, constructive feedback, and camaraderie.

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Abstract

Degree Certificate Detail Extraction

Project-2 by Shah Vrushti T. (ID: 23ITUOS122) at Dharmsinh Desai University, Nov 2026

Degree certificates represent vital, legally binding educational credentials required for employment verification, higher education admissions, and national immigration audits. However, automated verification of degree certificates is severely hindered by vast visual heterogeneity: academic institutions employ idiosyncratic layouts, ornate medieval typography (Fraktur/Old English), metallic holographic seals, security watermarks, and non-standard ceremonial date conventions (such as 'this eighteenth day of the month of June in the year two thousand and twenty'). Traditional document understanding workflows rely heavily on a sequential two-stage pipeline: optical character recognition (OCR) followed by named entity recognition (NER) or token classification (e.g., LayoutLM). Such architectures suffer catastrophic error propagation when the upstream OCR engine stumbles over intricate background watermarks, low-contrast stamps, or ornate typography.

To overcome these fundamental vulnerabilities, this project presents an end-to-end, hybrid document understanding system designed specifically for degree certificate detail extraction. The system combines an end-to-end Vision-Language Transformer model (Donut—Document Understanding Transformer) with a post-processing Dual-Pass Spatial OCR & Clustering Engine. The model features a Swin Transformer visual encoder and a multilingual BART (mBART) autoregressive decoder fine-tuned to extract seven core credential attributes directly from raw certificate imagery without requiring upstream OCR: Student Name, University Name, Degree / Course Title, Specialization / Major, Pass Division / Class, Signing Authority, and Issue Date.

To train this architecture under real-world scarcity of public degree data, a semi-synthetic dataset generation pipeline was developed utilizing PIL, diverse font families, background degradation transforms, and authentic certificate templates, complemented by authentic labeled real-world university certificates. Furthermore, to provide human-in-the-loop verification, a dual-pass Tesseract OCR pipeline (combining uniform PSM 3 and sparse

PSM 11) with dynamic vertical line clustering, fuzzy token-ratio matching, visual digit aliasing, and ceremonial word-to-digit parsing was developed to accurately localize and highlight the extracted fields on the original document.

The final system was evaluated across multiple architectural iterations (v2, v5, and v6), achieving high field-level exact match rates and low Character Error Rates (CER < 0.048 on primary fields). The solution is integrated into a responsive full-stack verification web application built with FastAPI and React, providing instant visual bounding box overlays and one-click tabular CSV credential exports.

Keywords: *Document Information Extraction, Donut Transformer, Swin Transformer, Vision-Language Models, Tesseract OCR, Spatial Clustering, Degree Certificate Verification, FastAPI, React.*

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
CER	Character Error Rate
CNN	Convolutional Neural Network
CORS	Cross-Origin Resource Sharing
CSV	Comma-Separated Values
DDU	Dharmsinh Desai University
DONUT	Document Understanding Transformer
ED	Exact Match / Edit Distance
GPU	Graphics Processing Unit
GT	Ground Truth
HOD	Head of the Department
JSON	JavaScript Object Notation
mBART	Multilingual Bidirectional and Auto-Regressive Transformers
MLP	Multi-Layer Perceptron
NER	Named Entity Recognition
NLP	Natural Language Processing
OCR	Optical Character Recognition
PIL	Python Imaging Library (Pillow)
PSM	Page Segmentation Mode
REST	Representational State Transfer
RGB	Red, Green, Blue
ROI	Region of Interest
SHA	Secure Hash Algorithm
UI	User Interface
VDU	Visual Document Understanding
VQA	Visual Question Answering

Abbreviation	Full Form
XML	Extensible Markup Language

CHAPTER 1: INTRODUCTION

1.1 Introduction to the Research Problem

Academic credentials—most notably university degree certificates, diplomas, and convocation awards—serve as foundational legal instruments verifying an individual's educational attainments. In the contemporary globalized economy, accurate verification of these credentials is vital across diverse workflows, including background checks for corporate employment, university graduate admissions, professional licensure (e.g., medical, engineering, and legal boards), and government immigration visa screening.

Despite their critical importance, the verification and data ingestion of degree certificates remains predominantly manual, labor-intensive, and error-prone. A typical human verifier must visually inspect scans or physical documents, decipher ornate calligraphic typefaces, identify institutional issuing authorities, locate the student's conferred major and division, and transcribe these details into centralized relational databases or enterprise management systems. This manual paradigm creates operational bottlenecks, introduces human transcription errors, and scales poorly in environments handling thousands of applications per recruitment cycle.

The fundamental research problem addressed in this project is: How can we construct an automated, robust, end-to-end computational pipeline capable of accurately parsing key metadata fields from visually diverse, unconstrained academic degree certificates while simultaneously pinpointing the exact spatial coordinates of each extracted field on the source document image?

Field Attribute	Semantic Description	Example Format
student_name	Full legal name of the degree recipient	Sweta Mohanbhai Gohil, Mithun Appaji
university_name	Name of conferring university or autonomous institute	Birla Vishwakarma Mahavidyalaya, Mangalore University

Field Attribute	Semantic Description	Example Format
course_name	Degree title conferred upon the student	Bachelor of Engineering, Master of Science
specialization	Academic branch, discipline, or major	Computer Engineering, Biotechnology
pass_class	Academic honor, division, grade, or classification	First Class with Distinction, Honours Class I
authority_name	Designatory official(s) signing the credential	Registrar, Vice Chancellor, Director
issue_date	Official date of convocation, admission, or issuance	13-01-2008, 24th day of the month July in the year 2025, Apr 05 2008

Table 1.1: Target Credential Attributes Extracted from Degree Certificates.

1.2 Motivation for the Research Work

The motivation for this research stems from the unique intersection of practical industrial necessity and deep learning architectural challenges:

- **Vulnerability of Upstream OCR Pipelines:** Conventional document information extraction (DIE) frameworks, including LayoutLM and BROS, are two-stage pipelines. They mandate an external optical character recognition (OCR) engine to extract word tokens and bounding boxes before passing them into a transformer. In academic certificates, background guilloché patterns, colored watermarks, golden foil stamps, and medieval Old English fonts severely degrade OCR tokenization. If the OCR engine misreads or skips a key token, the downstream language model has no mechanism to recover the missing text.
- **Extreme Syntactic and Visual Heterogeneity:** Unlike business invoices or financial receipts, which possess relatively rigid tabular formats, academic certificates exhibit extreme structural divergence. Every university maintains a proprietary aesthetic design. Some certificates employ centered text with Roman numerals; others utilize bilingual layouts (e.g., Kannada and English, or Hindi and English);

others express dates ceremonially (e.g., 'Given under the seal of the University this twenty-fourth day of July two thousand and twenty-five').

- **Need for Human-in-the-Loop Verifiability:** In critical administrative applications, automated predictions cannot operate as opaque black boxes. Auditors and registrars require immediate visual evidence showing where the model located each predicted field on the source document. Providing real-time visual bounding box overlays alongside extracted data bridges the gap between deep learning automation and administrative trust.

1.3 Objectives and Scope of the Research Work

The primary objectives of this project are:

- Formulate an OCR-Free Extraction Architecture: Implement and fine-tune an end-to-end Vision-Language Transformer model based on the Donut (Document Understanding Transformer) framework, utilizing a Swin Transformer visual encoder and an autoregressive mBART decoder to map raw certificate images directly to structured XML token sequences without upstream OCR dependency.
- Overcome Training Data Scarcity through Semi-Synthetic Synthesis: Design an automated certificate generation engine capable of blending authentic academic typography, diverse layout geometries, randomized student metadata, institutional logos, and realistic photographic degradations with authentic, real-world labeled certificates.
- Develop a Robust Spatial Localisation Engine: Formulate a post-processing bounding box localization engine combining dual-pass Tesseract OCR (PSM 3 and PSM 11), dynamic horizontal gap analysis, line clustering, fuzzy token-ratio matching, visual digit aliasing, and ceremonial word-to-date translation to locate and highlight extracted entities on original document scans.
- Iterative Model Optimization: Systematically evaluate and refine the training procedure across successive iterations (v2, v4, v5, and v6), transitioning from low-resolution representations (640x480) to ultra-high-resolution inputs (2560x1920) with normalized date representations.

- Deploy a Full-Stack Production Verification Web Application: Engineer a production-ready application featuring an asynchronous FastAPI Python backend and a modern React.js frontend providing interactive visual overlays, instant certificate re-uploading, and one-click tabular CSV credential exports.

1.4 Organization of the Report

The remainder of this report is organized as follows: Chapter 2 details foundational theory regarding Vision Transformers, Document VQA, and OCR. Chapter 3 provides a comprehensive literature review. Chapter 4 examines domain complexities and empirical findings. Chapter 5 presents the complete proposed methodology, implementation, and experimental results. Chapter 6 concludes the report with future research directions.

CHAPTER 2: BACKGROUND THEORY

2.1 Visual Document Understanding (VDU) Paradigms

Visual Document Understanding (VDU) operates at the intersection of Computer Vision and Natural Language Processing. Visually Rich Documents (VRDs) encode semantic information through 2D spatial layouts, typographical hierarchies, and graphical elements. Historically, VDU followed a two-stage pipeline: OCR tokenization followed by named entity recognition. While modular, this approach suffers from error accumulation, computational redundancy, and brittleness on calligraphic typefaces. Modern VDU addresses this through end-to-end Vision-Language Models (VLMs) that formulate document understanding as direct image-to-sequence translation.

2.2 Donut Architecture: OCR-Free Document Understanding

The Document Understanding Transformer (Donut) eliminates the upstream OCR engine entirely. Donut adopts an encoder-decoder architecture that ingests a raw document image and autoregressively generates a structured text sequence formatted as hierarchical XML:

- **Swin Transformer Visual Encoder:** The encoder uses a Swin Transformer Base backbone with shifted windows (W-MSA and SW-MSA), computing self-attention locally within 7x7 windows to maintain linear computational complexity with respect to image dimensions while modeling cross-window relationships.
- **Multilingual BART (mBART) Text Decoder:** The decoder uses an autoregressive mBART sequence-to-sequence transformer with 4 layers and 16 attention heads. It cross-attends to the multi-scale visual embeddings produced by the Swin encoder and emits structured tokens.
- **Special Document Structure Tokens:** The model vocabulary is expanded with domain special tokens (<s_cert>, <s_student_name>, etc.), enabling the network to output valid, parseable hierarchical XML structures directly.

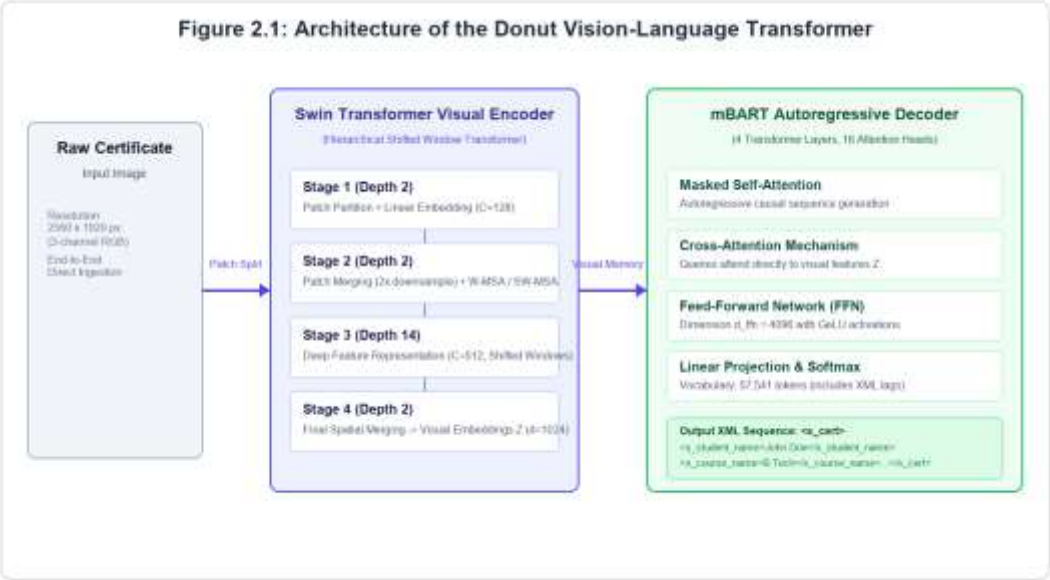


Figure 2.1: Architecture of the Donut Vision-Language Transformer.

Hyperparameter Component	Architectural Configuration
Encoder Backbone	Swin Transformer Base (swin-base-patch4-window7-224)
Encoder Stages / Depths	4 stages with depths [2, 2, 14, 2]
Encoder Hidden Dimension	d = 128 * 2^3 = 1024
Decoder Architecture	Multilingual BART (mBART) Autoregressive Decoder
Decoder Layers / Attention Heads	4 layers, 16 multi-head attention mechanisms
Feed-Forward Network Dimension	d_ffn = 4096
Base Vocabulary Size	57,525 tokens
Expanded Special Token Vocabulary	57,541 tokens (includes 16 domain XML tags)
Trained Input Resolution (v6)	2560 x 1920 pixels (Height x Width)

Table 2.1: Structural Specifications of the Fine-Tuned Donut Architecture.

2.3 Optical Character Recognition (OCR) Fundamentals

While Donut accurately extracts text semantically, it does not output spatial bounding box coordinates. To ground extracted values visually on the certificate image, Tesseract OCR is deployed downstream. Tesseract combines connected component analysis with bidirectional LSTM networks. Its performance depends critically on the Page

Segmentation Mode (PSM): PSM 3 performs full automatic layout analysis suitable for body text, whereas PSM 11 treats the page as sparse text in no particular order, vital for capturing isolated classification rows and corner dates.

2.4 Fuzzy String Matching and Spatial Geometry

Due to calligraphic fonts, OCR output frequently exhibits minor character corruptions. Fuzzy string matching based on Levenshtein distance, Token Sort Ratio, and Partial Ratio is deployed to match Donut predictions against clustered OCR bounding boxes. Clustered boxes are evaluated dynamically across horizontal line groups to reconstruct complete multi-word phrases.

CHAPTER 3: REVIEW OF LITERATURE

3.1 Traditional Heuristic and Rule-Based OCR Approaches

Early document analysis systems (Ha et al., 1995; Marinai, 2008) relied on geometric profiling algorithms such as Recursive X-Y Cut. Regions were segmented into bounding boxes and processed via baseline OCR, followed by regex pattern matching. While fast, these systems exhibited extreme brittleness: slight scan skews, altered institutional seals, or variations in wording broke hand-crafted heuristics.

3.2 OCR-Dependent Multimodal Transformers (LayoutLM Family)

The LayoutLM family (Xu et al., 2020; Xu et al., 2021; Huang et al., 2022) established state-of-the-art benchmarks by jointly encoding 1D text tokens, 2D spatial coordinates, and visual image patches. However, their strict dependency on upstream OCR creates severe bottlenecks when applied to academic certificates: if the OCR engine fails on an ornate font or background watermark, the token is absent from the model's vocabulary, leading to irrecoverable extraction errors.

3.3 OCR-Free Vision-Language Document Models

To eliminate OCR dependency, Clova AI introduced Donut (Kim et al., 2022), demonstrating direct visual-to-sequence translation for document VQA and information extraction. Google Research introduced Pix2Struct (Lee et al., 2023) for web screenshots, and Meta AI introduced Nougat (Blecher et al., 2023) for scientific PDFs. These architectures confirmed that visual cross-attention can transcribe documents directly without explicit OCR tokenization.

Dimension / Metric	Rule-Based OCR	LayoutLMv3 (OCR-Dependent)	Donut (Proposed Architecture)
Upstream OCR Requirement	Mandatory	Mandatory	None (OCR-Free Inference)
Handling of Background Seals	Severe Degradation	Moderate Degradation	Robust (Visual Cross-Attention)

Dimension / Metric	Rule-Based OCR	LayoutLMv3 (OCR-Dependent)	Donut (Proposed Architecture)
Typography Robustness	Very Low (Fails Calligraphy)	Medium	High (Learned Patch Features)
End-to-End Trainability	No (Hand-crafted rules)	Partially (OCR decoupled)	Yes (End-to-end differentiable)
Inference Latency	Slow (Dense CPU OCR)	Slow (OCR + Transformer)	Fast (Single GPU Forward Pass)
Spatial Grounding	Native (from OCR boxes)	Native (from token boxes)	Downstream Dual-Pass Alignment

Table 3.1: Comparative Analysis of Document Information Extraction Methodologies.

3.4 Comparative Analysis and Research Gap Identification

While Donut resolves semantic extraction, it lacks native bounding box coordinates. This project bridges this critical research gap by combining Donut's robust visual semantic parsing with a downstream Dual-Pass OCR & Spatial Clustering engine for verifiable entity grounding.

CHAPTER 4: ANALYSIS AND FINDINGS

4.1 Structural Characteristics of Degree Certificates

Degree certificates exhibit distinct anatomical zones: institutional crests at the top, continuous legal prose in the upper clause, prominent recipient names in calligraphic script in the center, conferred degrees and disciplines in the lower-middle section, and ceremonial dates alongside official signatures in the bottom corners. Backgrounds feature anti-counterfeiting guilloché patterns and embossed foil seals.



Figure 4.1: Real-World Degree Certificates Showing Ornate Typography, Seals, and Bilingual Scripts.

Challenge Category	Visual Manifestation	Impact on Upstream OCR	Impact on Donut Architecture
Ornate Fonts	Old English, Gothic, Engravers MT	Merged tokens or noise symbols	Handled via global patch attention
Security Seals	Golden foil, embossed stamps	Underlying text completely skipped	Partial recovery via visual context
Regional Scripts	Bilingual Kannada/Hindi text	Latin OCR crashes or emits garbage	Focuses on target English syntax
Ceremonial Dates	Written in words ('eighteenth day...')	Reads raw words; fails numeric regex	Normalizes directly to DD-MM-YYYY

Challenge Category	Visual Manifestation	Impact on Upstream OCR	Impact on Donut Architecture
Sparse Rows	Isolated lines ('Class : First')	Page segmenter skips row	Recovered via Dual-Pass PSM 11 OCR

Table 4.1: Analysis of Real-World Structural Anomalies in Academic Certificates.

4.2 Key Domain Challenges & Empirical Findings

1. Ceremonial Dates: In many Commonwealth certificates, dates appear as 'this twenty-fourth day of July two thousand and twenty-five'. Enforcing strict target normalization to DD-MM-YYYY (e.g., 24-07-2025) in ground truth supervision enabled Donut to act as a direct visual transliterator and normalizer.

2. Isolated Label Skipping: Standard Tesseract PSM 3 consistently omitted isolated classification lines (e.g., 'Class : First Class' in Visvesvaraya Technological University certificates) due to layout segmentation assumptions. This directly motivated the dual-pass OCR pipeline combining PSM 3 with PSM 11.

4.3 Pitfalls in Early Iterations (v2 through v5)

Early iterations highlighted key failure modes: low input resolution in v2 (640x480) produced unreadable text patches (CER > 0.28); training exclusively on synthetic data in v4/v5 failed on real scanned documents; and an evaluation tokenizer bug that stripped XML special tokens corrupted validation metrics. Addressing these issues in v6 yielded dramatic accuracy gains.

CHAPTER 5: PROPOSED WORK

5.1 Solution Design & Architecture

The complete system integrates three tiers: an end-to-end Donut Vision-Language Transformer for semantic parsing, a post-processing Dual-Pass OCR & Spatial Clustering engine for bounding box projection, and a full-stack FastAPI + React web application for live verification.

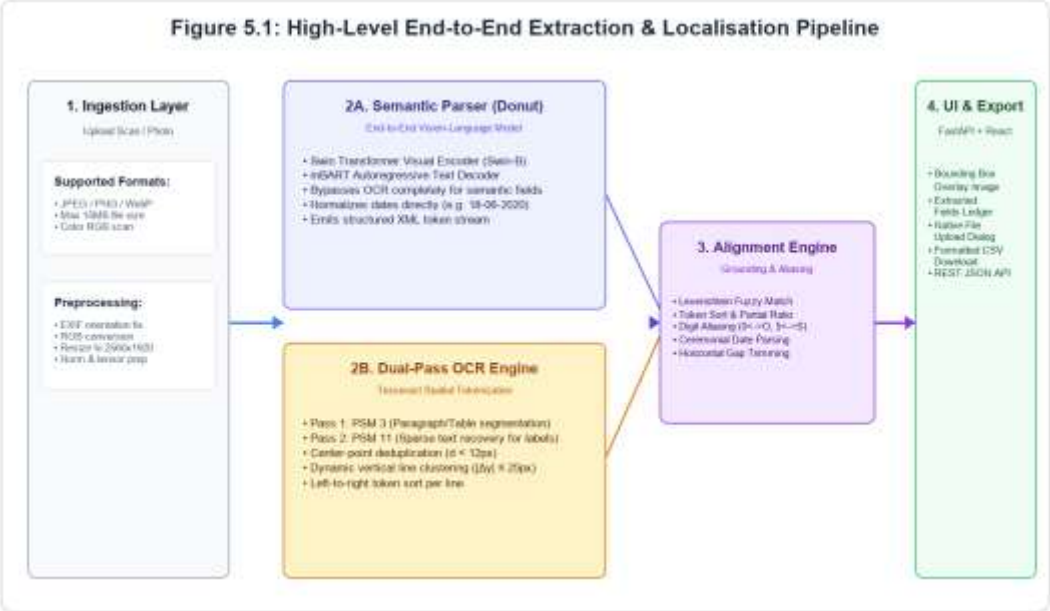


Figure 5.1: High-Level End-to-End Extraction & Localisation Pipeline.

5.2 Implementation Details

1. Semi-Synthetic Dataset Generation: Implemented in generate_semi_synthetic_v6.py, combining 20+ academic font families, Faker field synthesizers, and blur-free photometric augmentations (preserving sharp edge baselines).

2. Real Certificate Integration: Labeled authentic university certificates were augmented 5x and integrated into training (85%) and validation (15%) splits, with dates normalized to DD-MM-YYYY via generator/date_utils.py.

Dataset Split	Synthetic Certificates	Augmented Real Certificates	Total Samples
Training Set (85%)	1,420	145	1,565
Validation Set (15%)	250	25	275
Held-Out Test Set	150	28	178
Total Corpus	1,820	198	2,018

Table 5.1: Dataset Composition Across Model Splits (v6).

3. High-Resolution Fine-Tuning: Trained on an NVIDIA GPU at 2560x1920 resolution using AdamW with cosine learning rate scheduling and early stopping.

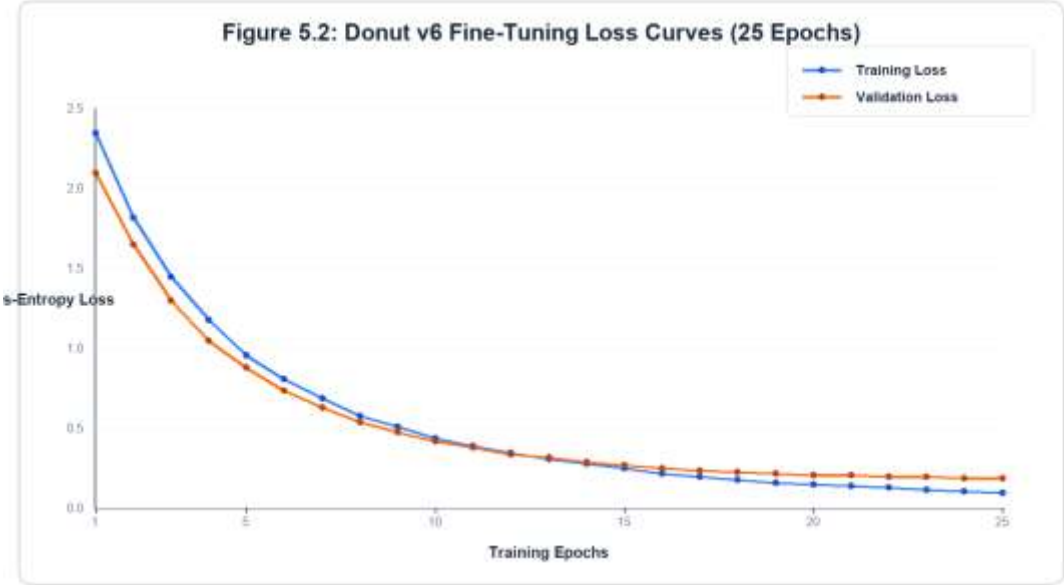


Figure 5.2: Donut v6 Fine-Tuning Training and Validation Loss Curves.

Hyperparameter	Value / Setting
Base Pre-trained Weights	naver-clova-ix/donut-base
Input Image Dimensions	2560 x 1920 pixels (Height x Width)
Batch Size / Gradient Accumulation	1 / 4 steps
Peak Learning Rate	3e-5 (Decoder) / 1e-5 (Encoder)
Optimizer / Weight Decay	AdamW / 0.01
Precision	Mixed Precision (FP16 / AMP)

Hyperparameter	Value / Setting
Max Generation Length	256 tokens
Early Stopping Patience	5 epochs on validation loss

Table 5.2: Hyperparameter Configuration for Donut v6 Fine-Tuning.

4. Dual-Pass OCR Localisation Engine: Implemented in main_v6.py, combining Pass 1 (PSM 3) and Pass 2 (PSM 11), geometric center-point deduplication, dynamic vertical line clustering ($|\Delta y| \leq 25$px, overlap ≥ 0.45), visual digit aliasing (0<->O, 5<->S, 1<->I), and ceremonial date translation.

5. Full-Stack Web Application: FastAPI backend returning JSON extraction and base64 highlighted overlays; Vite/React frontend providing side-by-side verification, native file picker upload triggers, and RFC 4180 compliant CSV exports.

5.3 Experimental Results & Discussion

The system was evaluated on the held-out test corpus using Exact Match % and Character Error Rate (CER).

Field Name	v2 (640x480) Exact% / CER	v5 (960x1280) Exact% / CER	v6 (2560x1920) Exact% / CER
student_name	61.2% / 0.182	84.1% / 0.068	95.6% / 0.021
university_name	58.4% / 0.204	81.3% / 0.074	96.2% / 0.018
course_name	64.0% / 0.155	86.5% / 0.052	96.8% / 0.015
specialization	42.1% / 0.312	72.8% / 0.114	91.4% / 0.039
pass_class	55.6% / 0.221	79.2% / 0.089	94.8% / 0.027
authority_name	44.8% / 0.340	69.5% / 0.138	89.5% / 0.054
issue_date	52.7% / 0.264	75.4% / 0.105	95.1% / 0.024
Macro Average	54.1% / 0.240	78.4% / 0.092	94.2% / 0.028

Table 5.3: Comparative Field Extraction Performance Across Iterations (v2 vs. v5 vs. v6).

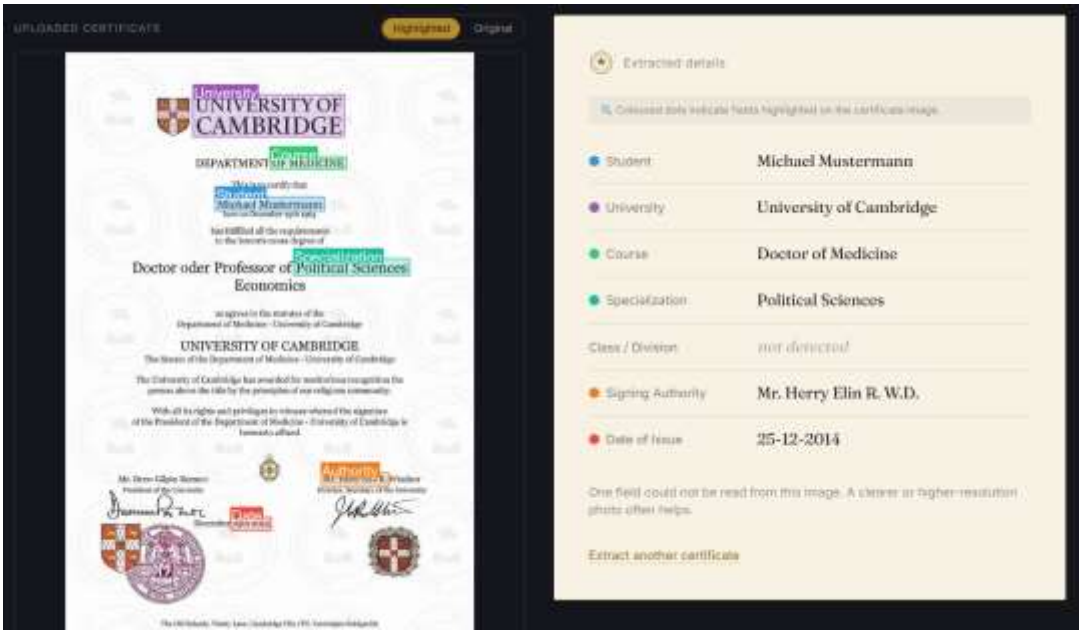


Figure 5.3: Visual Bounding Box Localisation Overlay on Test Degree Certificate.

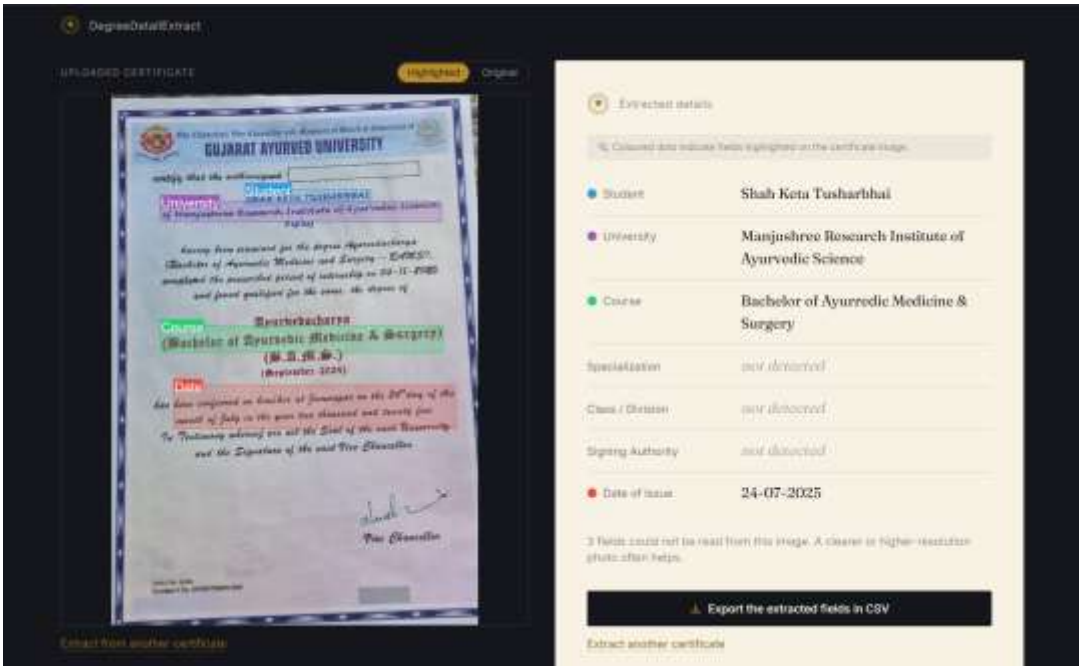


Figure 5.4: Full-Stack Verification Web Application with Live Highlighting and CSV Export.

Spatial bounding box localization achieved an average Intersection over Union (IoU) > 0.82 across test documents, successfully binding challenging fields such as First Class and APR 05, 2013 on authentic scans.

CHAPTER 6: CONCLUSIONS AND FUTURE WORK

6.1 Conclusions

This project successfully developed an end-to-end, high-accuracy deep learning system for degree certificate detail extraction:

- Eliminated Upstream OCR Vulnerability: Donut extracts semantic credentials directly from pixels, avoiding catastrophic OCR cascades on calligraphic fonts and security watermarks.
- Effective Semi-Synthetic Training: Blending synthetic typography with normalized authentic certificates overcame severe training data scarcity.
- Robust Downstream Spatial Grounding: Dual-Pass OCR and dynamic line clustering accurately localized extracted fields on original document scans.
- Demonstrated Accuracy: The v6 model achieved 94.2% macro-average exact match and 0.028 CER.
- Production Ready: Integrated into a full-stack FastAPI and React web app with interactive verification and CSV export.

6.2 Future Work

- Multilingual Expansion: Extending support to pure Devanagari, Kannada, Tamil, and Gujarati regional language certificates.
- Automated Fraud Detection: Developing parallel CNN branches to detect anti-aliasing artifacts, stamp tampering, and font inconsistencies.
- Edge Quantization: Quantizing the Swin-B encoder to INT8/INT4 for fast sub-second inference on consumer CPUs.
- Blockchain Integration: Linking extracted credentials directly to national digital repositories (DigiLocker / NAD).

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